

III. BACKGROUND

FDA issued guidance on electronic source data in clinical investigations (eSource guidance).¹¹ In the eSource guidance, FDA addresses attributes of **source data** used to fill predefined fields in an **electronic case report form (eCRF)** that would satisfy FDA's inspection, recordkeeping, and record retention requirements.¹² The guidance acknowledges that data entered into the sponsor's eCRF may be derived from a variety of sources, including EHRs.

In general, EHR systems are not under the control of FDA-regulated entities (e.g., sponsors, clinical investigators). In most instances, these systems belong to health care providers, health care organizations, and health care institutions. As provided in the eSource guidance, FDA does not intend to assess compliance of EHR systems with 21 CFR part 11.¹³ However, FDA's acceptance of data from clinical investigations for decision-making purposes depends on FDA's ability to verify the quality and integrity of the data during FDA inspections (see 21 CFR parts 312 and 812). This guidance clarifies FDA's expectations when EHRs are used as a source of data in clinical investigations.

Potential Advantages of the Use of EHRs in Clinical Investigations

With the widespread use of EHRs, there are opportunities to improve data accuracy and promote clinical trial efficiency when EHRs are used in clinical investigations. EHRs may enable clinical investigators and study personnel to have access to many types of data (e.g., clinical notes, physician orders, radiology, laboratory, and pharmacy records) that can be combined, aggregated, and analyzed. EHRs may have the potential to provide clinical investigators and study personnel access to real-time data for review and can facilitate post-trial follow-up on patients to assess long-term safety and effectiveness of medical products. In addition, there are opportunities for long-term follow up of large numbers of patients, which may be of particular importance in studies where the outcome of interest occurs rarely, such as in prophylaxis studies.

IV. INTEROPERABILITY AND INTEGRATION OF SYSTEMS

For the purposes of this guidance, **interoperability** refers to the ability of two or more products, technologies, or systems to exchange information and to use the information that has been exchanged without special effort on the part of the user. EHR and EDC systems may be noninteroperable, interoperable, or fully integrated, depending on supportive technologies and standards.

Noninteroperable systems, without the capability for electronic exchange of EHR data in clinical investigations, involve manual transcription of **data elements** from the EHR to the eCRF or to

¹¹ See the guidance for industry *Electronic Source Data in Clinical Investigations*.

¹² See footnote 5.

¹³ See footnote 11.

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the paper case report form, similar to the transcription performed with paper records. Such manual transcription procedures may introduce risks of data entry errors unless effective quality control systems are in place.

Interoperable systems allow electronic transmission of relevant EHR data to the EDC system. For example, data elements originating in an EHR (e.g., demographics, vital signs, laboratory data, medications) may automatically populate the eCRFs within an EDC system. In addition, an interoperable EHR and EDC system could provide access to additional patient information populated from other clinical information systems (e.g., radiology information systems, laboratory information systems). Interoperable systems may simplify data collection for a clinical investigation by enabling clinical investigators and study personnel to capture source data at the patient's point-of-care visit. Interoperable systems may also reduce errors in data transcription, allowing for the improvement in data accuracy and the quality and efficiency of the data collected in clinical investigations.

Fully integrated systems allow clinical investigators to enter research data directly into the EHR. This may involve, for example, use of research modules, use of research tabs built into the EHR system, or use of custom research fields within the EHR system for data that are entered for research purposes.

FDA encourages sponsors and clinical investigators to work with entities that control EHR systems, such as health care organizations, to use EHR and EDC systems that are interoperable or fully integrated. Moreover, diverse ownership of electronic systems and data may necessitate appropriate collaboration between the health care and clinical research communities. FDA encourages sponsors and health care organizations to work with EHR and EDC system vendors to further advance the interoperability and integration of these systems.

A. Data Standards

There may be practical challenges to the interoperability of EHR and EDC systems. These challenges may include the complex and diverse clinical data standards used by the health care and clinical research communities, which may hinder the exchange of information between different electronic systems. Many of these challenges are being addressed by the adoption of open data standards and through EHR data standardization requirements as part of the ONC Health IT Certification Program and ONC's Interoperability Standards Advisory.¹⁴ The data exchange between EHR and EDC systems should leverage the use of existing open data standards, when possible, while ensuring that the integrity and security of data are not compromised.

B. Structured and Unstructured Data

FDA encourages exchange of structured data (e.g., demographics, vital signs, laboratory data) between EHR and EDC systems so that data may be entered once at the point-of-care and used many times without manual re-entry or manual source data verification. Sponsors should ensure

¹⁴ The ONC Health IT Certification Program and their processes are discussed further in section V.A of this guidance. For more information on ONC's Interoperability Standards Advisory, see <https://www.healthit.gov/isa/>.

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that the structured data elements obtained from the EHR correspond with the protocol-defined data collection plan (e.g., time and method of measurement). In addition, for extraction of unstructured data, sponsors should consider the reliability and quality of unstructured EHR data and the appropriateness of using it as critical source data, such as study endpoints.

C. Validation

Sponsors should ensure that the interoperability of EHR and EDC systems (e.g., involving the automated electronic transmission of relevant EHR data to the EDC system) functions in the manner intended in a consistent and repeatable fashion and that the data are transmitted accurately, consistently, and completely. The sponsor's quality management plan (e.g., standard operating procedures, software development life cycle model, change control procedures) should address the interoperability of the EHR and EDC system and the automated electronic transmission of EHR data elements to the EDC system. Sponsors should ensure that software updates to the sponsor's EDC systems do not affect the integrity and security of EHR data transmitted to the sponsor's EDC systems. In addition, as part of the quality management plan, FDA encourages sponsors to periodically check a subset of the extracted data for accuracy, consistency, and completeness with the EHR source data and make appropriate changes to the interoperable system when problems with the automated data transfer are identified.

D. Data From Multiple EHR Systems

The EHR system at the clinical investigation site may be interoperable with multiple EHR systems from many different health care organizations or institutions that are not affiliated with the clinical investigation site. If data from multiple EHR systems from different health care organizations and institutions are integrated with EHR data at the clinical investigation site, data from another institution's EHR system may be used and transmitted to the sponsor's EDC system provided that data sharing agreements are in place.

V. BEST PRACTICES FOR USING EHRs IN CLINICAL INVESTIGATIONS

The use of EHRs as a source of data in clinical investigations may involve additional considerations, planning, and management as described in this section. Sponsors and clinical investigators should ensure that policies and processes for the use of EHRs at the clinical investigation site are in place and that there are appropriate security measures employed to protect the confidentiality and integrity of the study data.

Sponsors should also ensure that study monitors have suitable access to all relevant subject information pertaining to a clinical investigation, as appropriate. Such access must be described in the informed consent (see 21 CFR 50.25(a)(5)) (see section V.E). Furthermore, at any time during the course of a clinical investigation, sponsors should discuss with the relevant FDA review division any unique issues or challenges encountered relating to the data collection from the EHRs (see, e.g., 21 CFR 312.41).